
(a) dense W_{2d} (rows = output channels)
(b) vector step: same 4 of M columns for all V rows
(c) 2:4 step: 2 of the 4 kept, per row
V=4
M=8 columns
(d) packed V:N:M — what is stored and what Spatha reads
0 1 2 3
1 2 3 0
2 3 0 1
3 0 1 2
0 1 2 3
1 2 3 0
2 3 0 1
3 0 1 2
1 2 5 7 0 3 4 6
0 2 3 6 1 4 5 7
values (fp16)
$R \times K \cdot \frac{2}{M}$
2-bit index into the 4 kept columns
4 per (V-row block, M-group), 1 byte each
kept column positions 0..$M-1$:
density = $2/M$:
2:4 → 50%, 2:8 → 75%,
2:16 → 87.5%, 2:32 → 93.75% sparse.
small V: finer column choice (accuracy)
large V: bigger tiles (Spatha speed)